

PrimeFocus

May 2026



WHEN

May 15, 2026
Doors open at 7:00pm
Meeting starts at 7:30pm
Lecture starts at 8:00pm

WHERE

Unitarian Church
1893 North Vasco Road
Livermore, CA 945571
and via Zoom

TVS QR CODE



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THIS MONTH'S SPEAKER

Mission Control **Ben Heyman**

NASA's human spaceflight programs are global endeavor that bring together cutting edge engineering, international partnerships, and decades of operational expertise to explore low Earth orbit, the Moon, and beyond. This presentation offers an inside look at how Mission Control in Houston directs the daily operation of spacecraft such as the International Space Station (ISS), Boeing's CST-100 Starliner, and future lunar missions under the Artemis and Moon2Mars architecture.

Attendees will learn how Flight Controllers—highly trained specialists responsible for the health and performance of spacecraft systems—work around the clock to keep astronauts safe and missions running smoothly. Drawing from real mission experiences, Ben Heyman shares what it's like to serve as a SPARTAN and MPO Flight Controller, supporting ISS and capsule operations, commercial crew flights, and preparation for upcoming lunar exploration.



Ben Heyman is a NASA contractor working as a Station Power, Articulation, Thermal, and Analysis (SPARTAN) Front room Flight Controller (FCR) and Instructor for the International Space Station (ISS) Program, and Mechanical and Power Officer (MPO) Exploration Flight Controller providing technical expertise and insight into lunar exploration systems for the Moon to Mars (M2M) Program and Extravehicular Activity and Human Surface Mobility Program (EHP), with previous experience flying Boeing's CST-100 Starliner spacecraft.

Ben grew up in San Francisco, CA before attending Syracuse University in Upstate New York where he graduated with a Bachelor's Degree in Aerospace Engineering. In his free time, he enjoys exploring the Houston food scene, hanging with friends, directing a summer camp (Camp Go Beyond) for children with Inflammatory Bowel Disease in Livermore, CA, and traveling to colder climates.

NEWS AND NOTES

Meeting Dates

Club Meeting	Board Meeting	PrimeFocus Deadline
May 15	May 18	May 9
June 19	June 22	June 6
July 17	July 20	July 5

Money Matters

As of the last Treasurer's Report on 4/20/26, our club's account balance is \$53,174.05, this includes \$14,205.46 in the H2O Rebuild fund.

TVS Welcomes New Members

TVS welcomes new member Rajasi Joshi. Please say hello and chat with them during our meetings.

2026 TVS Club Star Party Schedule

Save the dates for the 2026 Club Star Parties. Del Valle star parties are also public outreach events. They are jointly hosted with the EBRPD and held at the Arroyo Staging Area (Coords: 37.6196638, -121.7528899). The public is invited for the first 1.5-2 hours, while club members can stay the remainder of the night.

No events currently scheduled for Del Valle.

Tesla Vintners star parties are open to only club members and their guests. These star parties end at midnight, but participants can leave earlier, should they wish.

June 6: Tesla Vintner's Star Party, 5143 Tesla Rd., Livermore. Set-up at 8:00pm, Observing 8:30-11:30pm.

August 8: Tesla Vintner's Star Party, 5143 Tesla Rd., Livermore. Set-up at 7:30pm, Observing 8:00-Midnight.

October 10: Tesla Vintner's Star Party, 5143 Tesla Rd., Livermore. Set-up at 6:30pm, Observing 7:00pm-11-30pm.

H2O Open House star parties are open to only club members and their guests. The open house ends at midnight, and all participants are encouraged to stay the duration. The drive to H2O takes about 1 hour, and the caravan leaves promptly from the corner of Mines and Tesla Rds. No gas stations are available on the route, so be prepared. Admission is \$3/car-bring exact change. H2O is a primitive site with two porta-potties. Bring water, food, and warm clothing, as needed. Red flashlights are to be used so observers can preserve their night vision.

September 5: H2O Open House, 5:00pm caravan to H2O PROMPTLY leaves the corner of Mines and Tesla Rds., Livermore. Observing until 11:30pm. Meeting times are tentative.

May 23: Public Star Party at Sunol Regional Wilderness, 1895 Geary Road, Sunol. Set up 7:30. Observing starts 8:00 pm, ends about 9:30 or 10 pm. Map to parking lot: <https://maps.app.goo.gl/sGxWWiah37SY7rhN8>

July 5: Outreach star party at Camp Arroyo, 5555 Arroyo Rd Livermore (The location is across the road from the Arroyo Road Staging Area. There is a locked gate at the entrance to the site, so you must let me know ahead of time if you plan to come, so we can make sure you have access.). Set-up 8:00pm, Observing from 8:30-11:00pm.

CALENDAR OF EVENTS



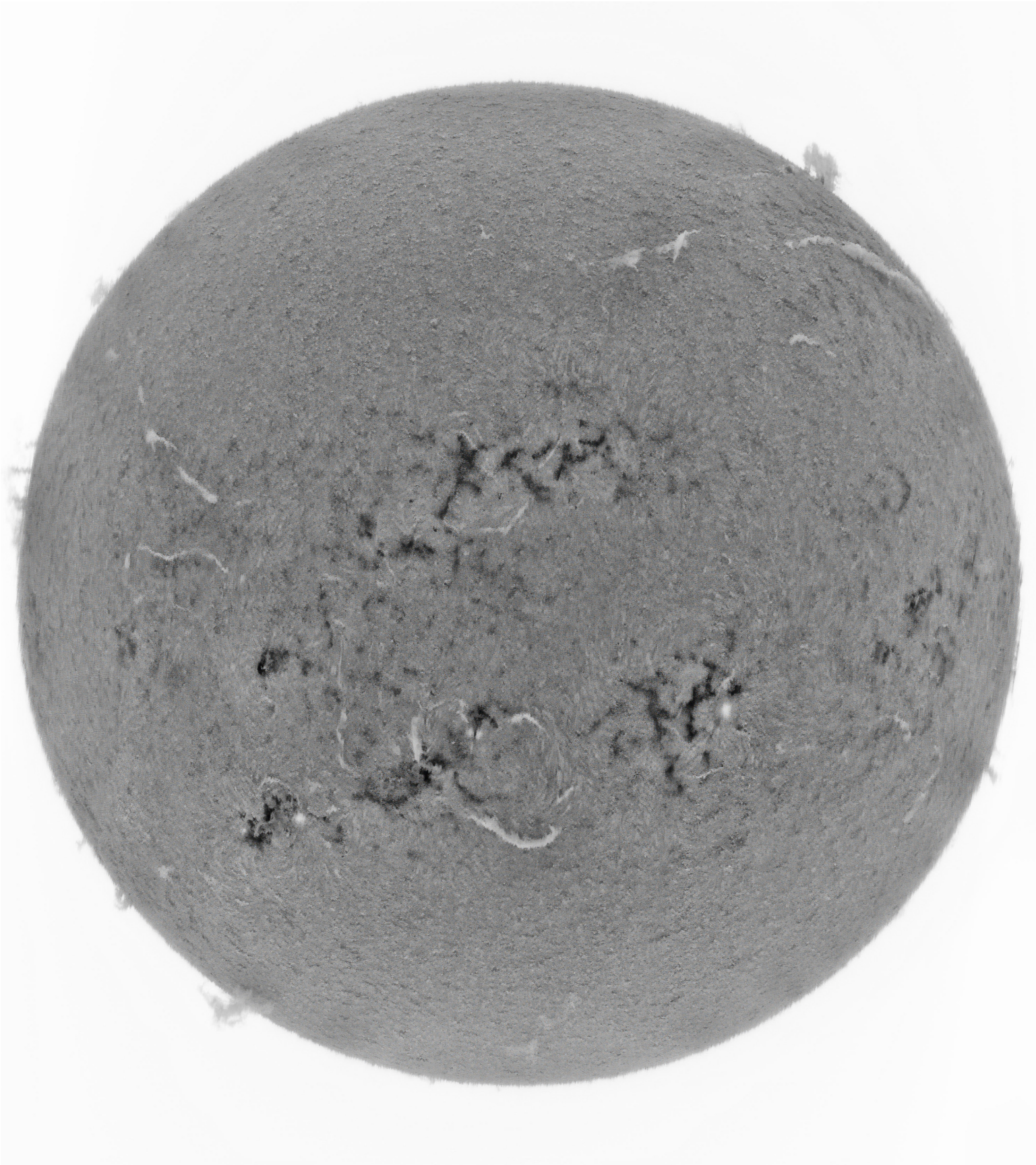
May 22, 23, 29,30, June 5, 6, 12,13, 19, 20, 7:30-10:30pm

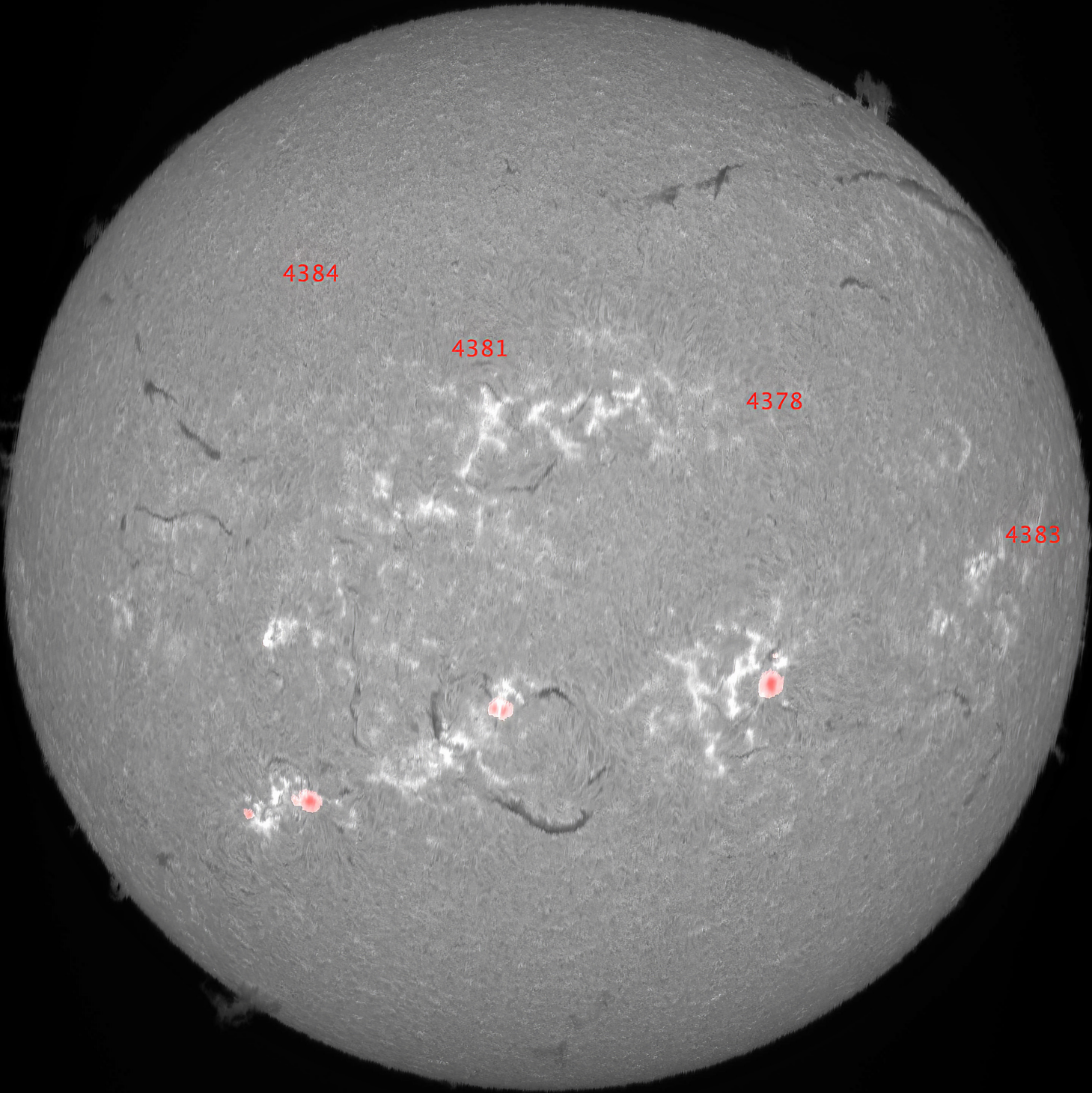
What	Free Telescope Viewing
Who	Chabot Staff
Where	Chabot Space and Science Center, 10000 Skyline Blvd. Oakland, CA 94619
Cost	Free

Join Chabot astronomers on the Observatory Deck for a free telescope viewing! Weather permitting, this is a chance to explore stars, planets and more through Chabot's historic telescopes. Chabot's three large historic telescopes offer a unique way to experience the awe and wonder of the Universe. Three observatory domes house the Center's 8-inch (Leah, 1883) and 20-inch (Rachel, 1916) refracting telescopes, along with a 36-inch reflecting telescope (Nellie, 2003).

Are the skies clear for viewing tonight? Viewing can be impacted by rain, clouds, humidity and other weather conditions. Conditions can be unique to Chabot because of its unique location in Joaquin Miller Park. Before your visit, check out the Weather Station to see the current conditions at Chabot.

For more information, see <https://chabotspace.org/events/events-listing/> <https://eastbayastro.org/chabot-telescope-status/>





WHATS UP

Adapted from Sky and Telescope

All times are Pacific Standard Time

May 2026

16	Sat	New Moon
18	Mon	At dusk look to the west-northwest to see Venus just 2° lower left of the Moon
20	Wed	Just after sunset look for the Moon lined up with Jupiter and Venus
22	Fri	at dusk look for Regulus shining just 3° upper left of the Moon
23	Sat	Moon is at first quarter
26	Tue	At dusk see the Moon just 5¼° right of Spica
30	Sat	At dusk look for the Moon rising just 2½° ahead of Antares
31	Sun	Full Moon

June 2026

7	Sun	At dusk facing west-northwest to see Venus about 4½° lower left of Pollux and Jupiter 2¼° to its left
8	Mon	Moon is at last quarter
9	Tue	At dusk Venus is just 1½° upper left of Jupiter
15	Mon	New Moon
16	Tue	At dusk looking west-northwest see the moon framed by Jupiter 4½° upper left, Pollux 4° upper right, Mercury 3° lower right, and Venus to the upper left
17	Wed	At dusk in the west, look for the Beehive Cluster (M44) near the Moon and Venus just 2½° lower right
19	Fri	At Dusk look for the Moon trailing Regulus by about 6°

OFFICERS AND VOLUNTEER POSITIONS

Officers	Volunteer Positions	Newsletter	Potluck Coordinator
President Eric Dueltgen president@trivalleystargazers.org	Astronomical League Rep. Don Dossa alrep@trivalleystargazers.org	Scott Schneider (Editor) newsletter@trivalleystargazers.org	OPEN potluck@trivalleystargazers.org
Vice-President Aris Pope vice_president@trivalleystargazers.org	Club Star Party Coordinator Tom Dossa coordinator@trivalleystargazers.org	Webmaster Swaroop Shere webmaster@trivalleystargazers.org	Program Coordinator Ron Kane programs@trivalleystargazers.org
Treasurer Ernest Epley treasurer@trivalleystargazers.org	Del Valle Coordinator Dave Wiltzius delvalle@trivalleystargazers.org	Night Sky Network Rep. Ross Gaunt nnsn@trivalleystargazers.org	Publicity and Fundraising OPEN publicity@trivalleystargazers.org
Secretary Dave Lackey secretary@trivalleystargazers.org	Historian OPEN historian@trivalleystargazers.org	H2O Observatory Director / Rebuild Chairman Chuck Grant H20@trivalleystargazers.org	Refreshment Coordinator OPEN
Past President Ron Kane past_president@trivalleystargazers.org	Librarian Ron Kane librarian@trivalleystargazers.org	Observing Program Coordinator Ron Kane awards@trivalleystargazers.org	Web and Email www.trivalleystargazers.org info@trivalleystargazers.org
	Loaner Scope Manager Ron Kane telescopes@trivalleystargazers.org	Outreach Coordinator Tom Dossa outreach@trivalleystargazers.org	TVS E-Group To Join the TVS E-Group just send an email to TVS at info@trivalleystargazers.org asking to join the group. Make sure you specify the email address you want to use to read and post to the group.

NAVIGATING THE NIGHT SKY FOR MAY

STAR CHART FOR MID-LATITUDES OF THE NORTHERN HEMISPHERE

What's Out Tonight?

May 2026 Sky Chart

OPTIMIZED FOR 1½ HOURS
AFTER SUNSET
but can be used after that for several more hours.

Instructions

Face North, South, East or West, then rotate the chart so your direction is at the bottom. Match the biggest stars on the chart to the brightest stars in the sky. The center of the chart is the top of the sky.

FREE

Visit: WhatsOutTonight.com

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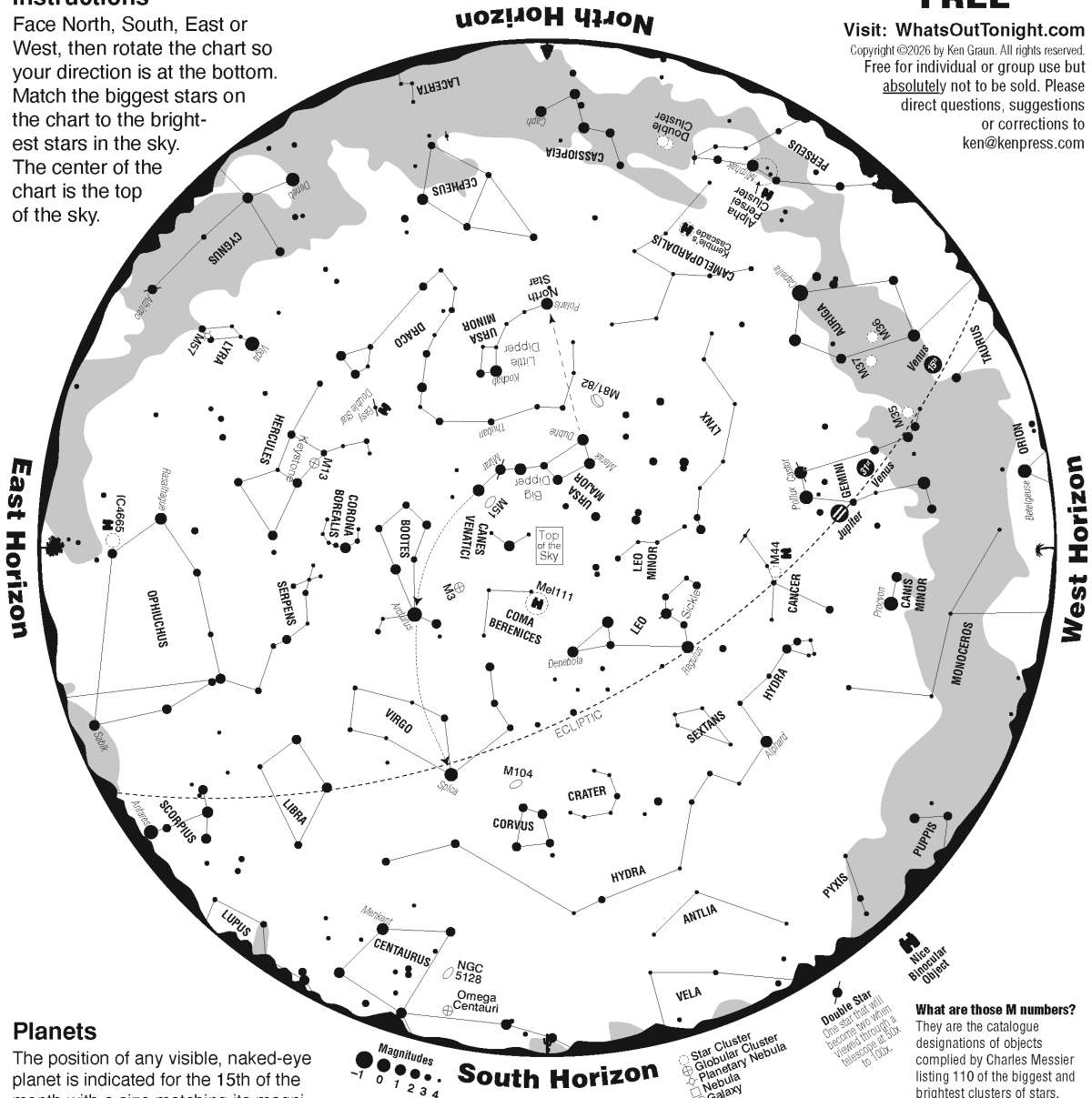
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direct questions, suggestions

or corrections to

ken@kenpress.com



Planets

The position of any visible, naked-eye planet is indicated for the 15th of the month with a size matching its magnitude. If the planet moves significantly during a month, other positions will be noted with dates. The **ECLIPTIC** is the path of the Sun through the sky but the planets and Moon move along it, too. It passes through the constellations of the zodiac.

May 2026 Planet Notes (15th of each month)

Venus, at magnitude -4.0 , sets in the west 2.5 hours after the Sun. **Mars**, at magnitude $+1.2$, in Pisces, rises in the east 1 hour before the Sun. **Jupiter**, at magnitude -1.9 , in Gemini, sets in the west about 12:30 AM. **Saturn**, at magnitude $+0.9$, in Cetus, rises in the east about 2 hours before the Sun.

Distances planets are from Earth the 15th of this month:

Venus: 126,000,000 miles, **Mars:** 206,000,000 miles

Jupiter: 534,000,000 miles, **Saturn:** 946,000,000 miles.

What are those M numbers?

They are the catalogue designations of objects compiled by Charles Messier listing 110 of the biggest and brightest clusters of stars, nebulae and galaxies.

May Notes

Near the top of the sky are two prominent constellations, LEO with its reverse question mark, the **Sickle**, punctuated by the kingly star, *Regulus*, and URSA MAJOR, containing the **Big Dipper** (the “backs” of Leo and the Big Dipper face each other). The handle of the dipper curves and points to the star *Arcturus* in BOOTES and continuing the curve, points to *Spica* in VIRGO. In front of Leo is CANCER containing the sprinkle of stars known as the Beehive. Less known and to Leo’s rear, is another sprinkle of stars in COMA BERENICES. Both sprinkles can be seen easily with binoculars and appear, to the eyes, as fuzzy patches in dark skies. At one time, the Beehive was thought of as the whiskers of Leo with the sprinkle of stars in Coma Berenices as the end of its tail.

SELECTED Clusters, Nebulae, Galaxies +

ly = Light year, a unit of distance. 1 ly = 6 trillion miles.
Our Moon (and Sun) spans 30' (30 arc minutes) or 1/2°.

Castor Double Star. Favorite double star. Need a telescope with 50x to 100x to see Castor separate into two stars. Magnitudes of two stars are 1.9 and 3.0. In GEMINI.

Coma Cluster. Sprinkle of 40+ stars. Appears as a faint haze in dark skies. Spans 4.5°. In COMA BERENICES

Kemble’s Cascade. Twenty magnitude 8 stars in a 2.5° line. The middle star is a “bright” 5th mag. In CAMELOPARDALIS.

M13. Globular Cluster. Distance: 21,000 ly / Diameter: 104 ly / Mag 5.8 / Spans 17'. Contains 500,000 stars. In HERCULES.

M35. Cluster. Distance: 3000 ly / Diameter: 24 ly / Mag 5 / Spans 28' / 200 stars. In GEMINI.

M36. Cluster. Distance: 3,700 ly / Diameter: 13 ly / Mag 6.0 / Spans 12' / 60 stars. Try with binoculars, too. In AURIGA.

M37. Cluster. Distance: 4,200 ly / Diameter: 29 ly / Mag 5.6 / Spans 24' / 150 stars. Try with binoculars, too. In AURIGA.

M44. Beehive Cluster. Distance: 610 ly / Diameter: 16 ly / Mag 3 / Spans 1.6° / 50 stars. In CANCER.

M51. Whirlpool Galaxy. Distance: 37 million ly / Diameter: 118,000 ly / Mag 8.1 / Spans 11'. In CANES VENATICI.

M81/82. Bode’s Galaxies. Visible in many light polluted skies. M82 shaped like a cigar. Mags 6.8/8.1. In URSA MAJOR.

M104. Sombrero Galaxy. Distance: 48 million ly / Diameter: 126,000 ly / Mag 8.1 / Spans 9'. In VIRGO.

Mizar. Two stars with good eyes or binoculars. Three stars with a telescope at 50x. Located in the handle of the Big Dipper.

Omega Centauri. Largest Globular Cluster! Distance: 17,000 ly / Diameter: 5 ly / Mag 3.7 / Spans 30'+. Contains 1 million stars. Looks like a faint star. Best in scope. In CENTAURUS.

Observing Tips for above Objects

If possible, observe at a dark location and when the Moon is not bright. A bright Moon will make it more difficult to see the stars and impossible to see clusters, nebulae and galaxies. Only a small telescope at lower magnifications, around 50x, is required to see the objects listed above. The planets and Moon are best observed with a telescope around 50x or more! To get a feel for the size of objects, the Moon extends 30' (30 arc minutes). The binocular objects are best with binoculars because these objects are large in size—telescopes have too much magnification.

Meteor Showers

The **Eta AQUARIDS** peak around **May 5** with 60 meteors/hour.

Brightest Stars

Antares. In SCORPIUS. Magnitude +1.1. Distance: 604 ly. Diameter: 300 times the Sun’s. Red Supergiant.

Arcturus. In BOOTES. Magnitude –0.04. Distance: 37 ly. Diameter: 26 times the Sun’s. It’s an Orange Giant.

Capella. In AURIGA. Magnitude +0.1. Distance: 42 ly. Diameter: 15 times the Sun’s. It’s actually 4 orbiting stars.

Castor. In GEMINI. Magnitude +1.6. Distance: 52 ly. Favorite double star that is twice the diameter of the Sun.

Deneb. In CYGNUS. Magnitude +1.3. Distance: about 1500 ly. Burning fast & might supernova in a few million years.

Polaris. In URSA MINOR. Magnitude +2. Distance: 431 ly. 2,400 times brighter than the Sun. Supergiant star.

Pollux. In GEMINI. Magnitude +1.2. Distance: 34 ly. Diameter is 8.8 times the Sun’s & 46 times brighter.

Procyon. In CANIS MINOR. Magnitude +0.4. Distance: 11.4 ly. Diameter is 2 times the Sun’s & 7.5 times brighter.

Regulus. In Leo. Magnitude +1.4. Distance: 78 ly. Diameter: 3.5 times the Sun’s & 140 times brighter.

Spica. In VIRGO. Magnitude +1.1. Distance: 262 ly. Actually two close stars revolving around each other in 4 days.

Vega. In LYRA. Magnitude +0.0. Distance: 25 ly. Rotates on axis once ever 12.5 hours. Mass is about 2.3 times our Sun.

May Mythology

FOR THE CENTRAL CONSTELLATIONS, NORTH TO SOUTH

Aras and his beautiful mother, Callisto were turned into the Little and Big Bears, **URSA MINOR** and **MAJOR** because of jealous Juno, wife of promiscuous Jupiter.

During a war between the Titans and Olympians, **DRACO**, the Dragon was flung to the North and frozen in place by the cold.

Regulus, the brightest star in **LEO**, the Lion has several meanings including regal, king and mighty. Before him is **CANCER**, the Crab sent to prevent **HERCULES** from killing the nine-headed **HYDRA** as one of his twelve labors toward a virtuous life.

CORVUS was a bird placed in the heavens on Hydra’s back by Apollo for being slow for bringing water and lying about his tardiness. **CRATER** represents the container of water that is always out of reach of Corvus.

When **VIRGO**, the Virgin is in the night sky, crops grow. The growing season ends when, in the early evening, she sets on the western horizon.

CANES VENATICI are the Hunting Dogs of **BOOTES** who is sometimes seen as a Ploughman. **CORONA BOREALIS** is the crown of Bacchus, the god of wine.

May Moon Phases

- **Full Moon.** Friday, **May 1**, 12:23 pm, CDT
- ◐ **Third or Last Quarter.** Saturday, **May 9**, 4:10 pm, CDT
- **New Moon.** Saturday, **May 16**, 3:01 pm, CDT
- ◑ **First Quarter.** Saturday, **May 23**, 6:10 am, CDT
- **Full Moon.** Sunday, **May 31**, 3:45 am, CDT

What’s Out Tonight?

May 2026 Sky Chart

Visit: WhatsOutTonight.com

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What's Out Tonight? • Sky Chart Supplement

Clusters, Nebulae & Galaxies

An **Open Cluster** is a group of several to hundreds of stars that were born out of the same nebula cloud. A group often forms a pretty pattern. The Pleiades and Praesepe are great examples. Open clusters reside in our Milky Way Galaxy. Our Sun is no longer in its group.

Globular Clusters look like fuzzy balls because they contain tens of thousands stars held together by their mutual gravity. There are 150+ of them that surround our Milky Way Galaxy like a halo—about 30 brighter ones visible from the northern hemisphere. M22 in SAGITTARIUS and M13 in HERCULES are favorites.

A **Planetary Nebula** is an old term that has nothing to do with the planets. Instead, it is a round or symmetrical nebula that is the shed atmosphere of a dying star. At its center is a white dwarf star. When our Sun dies, it will create a planetary nebula. These objects have diameters of a few light years and are located in our galaxy. The Ring Nebula, M57, in LYRA is a favorite.

A **Nebula** is a giant hydrogen gas cloud that is located in our galaxy. Within these clouds, concentrations of gas can occur and gravitationally condense to form stars and accompanying planets. A set of stars created by a nebula is known as an Open Cluster. The Orion Nebula, M42 is a favorite.

Galaxies contain billions of stars. All galaxies are beyond our Milky Way Galaxy, where our Sun resides. When you are observing a galaxy, you are looking through our galaxy into the true depths of the universe. The Andromeda Galaxy, M31 can be seen with the naked eye.

Double Stars

A Double Star is a star that looks like one star but when magnified sufficiently (from 6x to 200x+), it separates into two or more stars. Some are very pretty because of contrasting colors. *Castor* in GEMINI is a favorite and *Albireo* in CYGNUS is very much enjoyed for its blue & gold colors.

Moon

Starting from New Moon, the Moon cycles through phases every 29 days, 12 hours, 44 minutes, 3 seconds. It is 2,160 miles in diameter and averages 239,000 miles from Earth. A New Moon is not visible in the sky because the Moon is positioned very close to the Sun. Solar eclipses occur at New Moon. The best time to observe the Moon is during a phase because the craters appear their sharpest near the terminator, the line that separates the lighted side (day side) from the dark side (night side).

Cycle of Moon Phases



At arm's length...



One thumb width is 4 Moon diameters.



Orion's height is one hand span.



The width of a fist spans the Big Dipper's bowl.

Planets

The planets are best observed with a telescope using magnifications from 50x to 250x but 100x+ is dependent on a steady night sky. The five naked-eye planets are Mercury, Venus, Mars, Jupiter and Saturn. Venus is the brightest "star" and hangs close to the Sun, so you see it for a short time in the west after sunset or in the east before sunrise. Jupiter can be out all night and always outshines any star. Everyone enjoys its cloud bands and four Galilean moons that are easily visible at 50x and possible with well-focused binoculars. Saturn has its beautiful rings. Mars gets close to Earth every 2 years at which time it is very bright. This is the best time to observe it but you need higher magnifications around 150x+ to more easily see the surface coloration.

	Diameter In Miles	Rotation Its Day	Distance from Sun In Miles	Revolution Its Year
SUN	865,000	30 days	—	—
MERCURY	3,032	59 days	36,000,000	88 days
VENUS	7,521	243 days	67,000,000	225 days
EARTH	7,926	24 hours	93,000,000	365 days
MARS	4,228	24.6 hours	142,000,000	687 days
JUPITER	88,844	9.8 hours	484,000,000	11.8 years
SATURN	74,900	10.2 hours	887,000,000	29 years
URANUS	31,764	17.9 hours	1,800,000,000	84 years
NEPTUNE	30,777	19.2 hours	2,800,000,000	164 years
PLUTO*	1,433	6.4 days	3,700,000,000	248 years

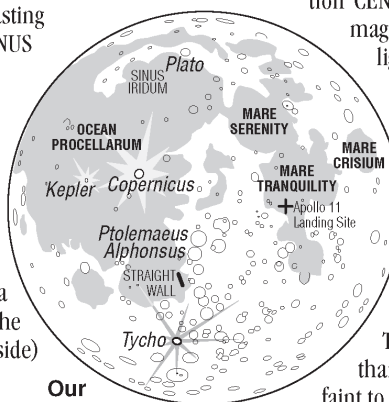
*Demoted to a Dwarf Planet in 2006 because it is part of an outer debris field.

Light Year (ly) & Nearest Stars

A Light Year (ly) is a *unit of length* and is equal to the distance light travels in 1 year. Since light moves at the speed of 186,282 miles a second, 1 light year is nearly 6 trillion miles long. The closest star visible to the naked eye is the southern hemisphere *Alpha (α) Centauri* or *Rigel Kentaurus* in the constellation CENTAURUS. It shines brightly at magnitude -0.27 and is just 4.4 light years away. The very closest star is *Proxima* in CENTAURUS at just 4.22 ly away but at magnitude $+11$, it's too faint to see with the eyes.

The second closest star visible to the naked eye is *Sirius* at 8.6 ly followed by *Epsilon (ε) Eridani* at 10.5 ly and *Procyon* at 11.4 ly.

There are several stars closer than these three but they are too faint to be seen with the naked eye.



Our Moon

What's Out Tonight? Sky Charts

Visit: WhatsOutTonight.com

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What's Out Tonight?

More Celestial Tidbit about . . .

...the M-number Designations.

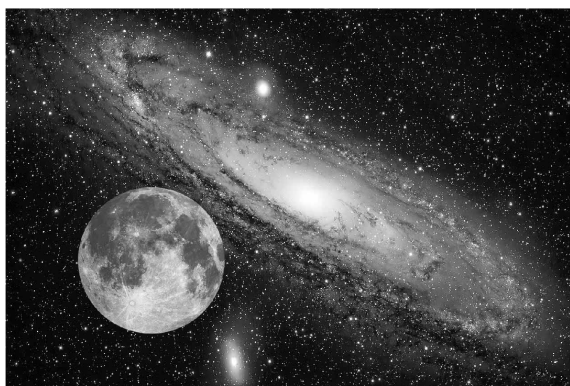
The Frenchman, Charles Messier, from Paris, during the mid-1700s compiled the very first catalogue of what is known today as Deep Sky Objects, which are clusters of stars, nebulae and galaxies. As the leading observational astronomer of his time he catalogued these non-stellar objects because such a catalogue did not exist, discovering a total of 110. Since he was the first to compile such a catalogue, his listing included most of the biggest and brightest objects that can be seen from the northern hemisphere—he used the equivalent of 4-inch diameter telescopes. Even though there are other, more comprehensive catalogues, Messier's is still used today but with an added M placed in front of his numbering. His catalogue serves beginning amateurs well as a first foray into observing Deep Sky Objects.

...the Constellations.

There are a total of 88 constellations in the sky. Each constellation has a boundary. If you include the boundary areas, CRUX (visible from the southern hemisphere) occupies the least area and HYDRA the most.

SERPENS is the only constellation that is split between two parts of the sky—on opposite sides of OPHIUCHUS. The eastern part is known as SERPENS Cauda (Cauda means tail) and the western part is known as SERPENS Caput (Caput means head).

In the northern hemisphere, we use the same constellations described by (visible to) the ancient Greeks, but a few have been added since then.



The Moon and the Andromeda Galaxy never get this close to one another in the sky, but this photo shows their comparative size. If Andromeda was very bright, as pictured, it would take 6 Moons to span its length. This image of the galaxy does not show the extremities of the galaxy. Many galaxies and nebulae span an area in the sky greater than that of the Moon, but because they are very faint, this evades attention and these faint objects do seem smaller than the Moon when viewed in a telescope—but they are not!

...the Stars.

The names of many stars are derived from Arabic. About 5,000 stars can be seen with the naked eyes—this includes both hemispheres. With a good dark sky, many people can see to about magnitude +6. Stars twinkle because of turbulence in the atmosphere and twinkle most when low in the sky because their pinpoint light gets easily refracted from different and moving layers of the atmosphere.

Greek Star Designations. Many of the “brighter” stars in the northern hemisphere have an associated lowercase Greek letter designation. This starts anew with each constellation and usually with the brightest star designated alpha (α) and so on down the line. Of course, many of these designated stars are named, too. These Greek letter designations were assigned by Johann Bayer in the early 1600s.

Brightest Naked-eye Stars		
	CONSTELLATION	MAG.
Sun		-26.73
[Full Moon]		-12.7
[Venus] average brightness		-4.1
[Jupiter] average brightness		-2.2
Sirius	Canis Major	-1.46
Canopus*	Carina	-0.72
Arcturus	Bootes	-0.04
Rigel Kent*	Centaurus	-0.27
[Saturn] average brightness		0.0
Vega	Lyra	+0.03
Capella	Auriga	+0.08
Rigel	Orion	+0.11
[Mercury] average brightness		+0.23
Procyon	Canis Minor	+0.34
Achernar*	Eridanus	+0.50
[Mars] average brightness		+1.0

*Southern Hemisphere

...the Milky Way.

The Milky Way Band is the faint glow from most stars that make up our Milky Way Galaxy. Our Sun and its accompanying solar system reside in the Milky Way Galaxy, which consists of hundreds of billions of stars all gravitationally bound to one another. The shape of our galaxy is that of a flat dish with a central bulge and having an overall diameter of around 100,000 light years. The direction to the center of our galaxy is between the constellations SAGITTARIUS and SCORPIUS, which happens to be the brightest part

of the Milky Way Band. The Andromeda Galaxy is in the constellation ANDROMEDA, is a companion galaxy to ours, and can just be seen with the naked eyes under dark skies. If you could see its full extent, it would span 6 Moon diameters—there are plenty of pictures on the internet of this beautiful galaxy. Our galaxy and Andromeda are part of the Local Group of about 30 galaxies—most are on the smaller size. There are a few hundred billion galaxies in the Universe.

Star Charts

- ★ 11-inch Diameter, ISBN 1-928771-03-3
- ★ 16-inch Diameter, ISBN 1-928771-01-7
- ★ Equatorial, ISBN 1-928771-77-7
- ★ Southern Hemisphere, ISBN 1-928771-11-4
- ★ Beginner's, ISBN 1-928771-99-8

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Sky Charts

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Tri-Valley Stargazers
P.O. Box 2476
Livermore, CA 94551
www.trivalleystargazers.org

Tri-Valley Stargazers Membership Application

Contact information:

Name: _____ Phone: _____

Street Address: _____

City, State, Zip: _____

Email Address: _____

Status (select one): _____ New member _____ Renewing or returning member

Membership category (select one): Membership term is for one calendar year, January through December.

_____ Student member (\$10). Must be a full-time high-school or college student.

_____ Regular member (\$30).

Hidden Hill Observatory Access (optional): Must be 18 or older.

_____ One-time key deposit (\$20). This is a refundable deposit for a key to H2O. New key holders must first hear an orientation lecture and sign a usage agreement form before using the observing site.

_____ Annual access fee (\$10). You must also be a key holder to access the site.

Donation (optional):

_____ Tax-deductible contribution to Tri-Valley Stargazers

Total enclosed: \$ _____

Member agrees to hold Tri-Valley Stargazers, and any cooperating organizations or landowners, harmless from all claims of liability for any injury or loss sustained at a TVS function. TVS will not share information with anyone except as detailed in our Privacy Policy (<http://www.trivalleystargazers.org/privacy.shtml>).

Mail this completed form along with a check to: Tri-Valley Stargazers, P.O. Box 2476, Livermore, CA 94551.